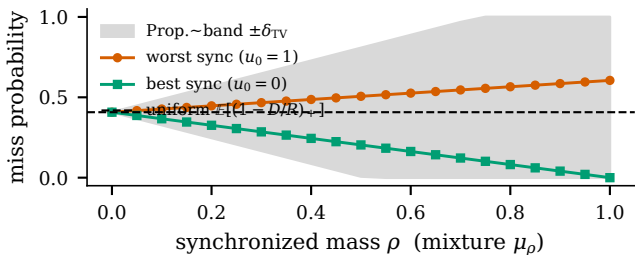


(a) perturbation bound, $R = 5$



(b) worst-case ceiling across R

